dsbda-as6

March 25, 2025

[1]:
```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score,␣
↪recall_score
from sklearn import datasets
```

[13]:
```python
# Load the dataset
iris = datasets.load_iris()
X = iris.data
y = iris.target
```

[3]:
```python
# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)
```

[4]:
```python
# Create and train the Naïve Bayes model
nb_model = GaussianNB()
nb_model.fit(X_train, y_train)
```

[4]:
```
GaussianNB()
```

[5]:
```python
# Make predictions
y_pred = nb_model.predict(X_test)
print(y_pred)
```

```
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
```

[6]:
```python
#Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
```

[7]:
```python
# Compute metrics
accuracy = accuracy_score(y_test, y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test, y_pred, average='weighted')
recall = recall_score(y_test, y_pred, average='weighted')
```

[8]:
```python
# Display results
print("Confusion Matrix:")
print(cm)
print(f"Accuracy: {accuracy:.4f}")
print(f"Error Rate: {error_rate:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
```

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
Accuracy: 1.0000
Error Rate: 0.0000
Precision: 1.0000
Recall: 1.0000
```

[ ]:
```python

```